

A025 · Deciphering Fetal Endothelial Cell Programs to Enhance Vascular Maturation in Human Organoid Models

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The development of the vascular system occurs through a coordinated sequence of events, including the generation of endothelial progenitor cells, their differentiation into endothelial cells (ECs), the formation of endothelial-lined blood islands in the yolk sac, the establishment of a primitive vascular plexus, and subsequent migration of vascular cells into the embryo, where extensive vascular networks are formed and remodeled. Despite advances in organoid technology, a fully functional vascular system that faithfully recapitulates human vascular development has not yet been achieved in vitro. Here, we compared the transcriptomic profiles of fetal endothelial cells (fECs) with those of umbilical vein endothelial cells (UVECs) and adult endothelial cells, two endothelial sources commonly used for vascularized organoid generation, to identify unique molecular features of fECs that are absent or reduced in these alternative EC populations. Transcriptomic analysis revealed that mitochondrial oxidative phosphorylation pathways, including genes such as ATP5A1 and ATP5B, and tRNA aminoacylation pathways, including TARS1 and SARS1, were among the most prominent features distinguishing fECs. Furthermore, fECs exhibited increased expression of cell adhesion-related genes, including CCN2 and CEMIP2, suggesting enhanced capacities for vascular organization and tissue integration. These findings provide insights into the molecular characteristics of developmentally relevant endothelial cells and may guide the generation of more physiologically functional vascularized human organoids. Improved vascular organoid models could facilitate advances in developmental biology, drug discovery, disease modeling, and precision medicine.